

How to setup IFIFF

Getting the IFDIFF code files

Method A: If MATLAB is installed locally:

1. Download `ifdiff-workshop2025.zip` from the HeiBox: <https://t1p.de/ifdiff2025materials>.
2. Unzip the file and place the resulting folder in a location of your choosing on your PC (e.g. Desktop).
3. Open MATLAB and navigate to your IFDIFF folder.

Method B: If you do not have a local MATLAB installation:

1. Open the IFDIFF repository in MATLAB Online by entering this link in your web browser: <https://matlab.mathworks.com/open/github/v1?repo=andreassommer/ifdiff>.
2. If asked to, login to your existing MathWorks account or create one. You do not need to possess a MATLAB license; creating a MathWorks account is completely free.
3. A popup will appear prompting you to save the IFDIFF repository in your MATLAB drive. Select a suitable location, check the checkbox and click on the “Save and Open” button.

Method C: Fallback if Method B does not work:

1. Download `ifdiff-workshop2025.zip` from the HeiBox: <https://t1p.de/ifdiff2025materials>.
2. Unzip the file and place the resulting folder in a location of your choosing on your PC (e.g. Desktop).
3. Open the MATLAB page in your web browser: <https://matlab.mathworks.com/>.
4. If you are not already signed in, click sign in and sign into your MathWorks account or create an account, if you do not already have one.
5. Click the “Open MATLAB Online” button.
6. Right click the files tab on the left side of the screen and select “Upload Folders” from the drop-down menu.
7. Select the IFDIFF folder you unzipped in step 2.
8. Your browser may show you a warning asking you to confirm if you want to upload all files in the folder. Click the upload button in the browser pop-up.
9. Navigate into the uploaded IFDIFF folder in your files tab.

Initializing the IFDIFF project (only once)

1. Make sure that you have opened the IFDIFF project folder in MATLAB. If not, then use the files tab to navigate to the correct folder, which should

contain many function files and other directories (e.g. a folder named `ComputeSensitivities`).

2. Click into the Command Window in MATLAB, type in the command `make_mtreeplus` and hit enter to run the command.
3. A text prompt should appear in your Command Window asking you whether to proceed. Enter “y” (without quotations) into the Command Window and hit enter to proceed.
4. Wait for the script to finish running. If the script produced an error, ask one of the IFDIFF Team members for assistance.

Setup the MATLAB search path to access IFDIFF functions (once per MATLAB session)

1. Note: You will need to repeat this step once per MATLAB session (e.g. if you need to restart MATLAB at some point).
2. Click into the Command Window in MATLAB, type in the command `initPaths` and hit enter to run the command.